

Harvard University
Physics 141 Pset 3

Due October 3, 2025

1. In the following problem, you will explore the statistics of the two-alternative forced choice (2AFC) task in Tinsley et al., 2016 (we studied this paper in class on Thursday). The paper reports $n = 2420$ single-photon events that passed post-selection, with an average proportion of correct response to be $\hat{p} = 0.516 \pm 0.010$. They also report $n = 242$ high-confidence responses, for which the proportion of correct responses was $\hat{p} = 0.60 \pm 0.03$. Note that in this notation $\hat{p}$ denotes the observed number of correct responses in a dataset and is an estimate of the probability of a correct response p . We are going to study, via a simple model, whether this data provides evidence that the true performance is above chance.

If each trial is independent with a probability of p to be correct, then the number of correct responses X after n trials follows a binomial distribution. The sample proportion is defined as $\hat{p} = \frac{X}{n}$. For large n , $\hat{p}$ follows a normal distribution with mean p and variance $p(1 - p)/n$.

- (a) Explain why the baseline probability of a correct response for the 2AFC task in Tinsley et al., 2016 is $p_0 = 0.5$.
- (b) Under a null hypothesis H_0 , we assume that the probability of a correct response is $p_0 = 0.5$. For the $n = 2420$ trials and under this null hypothesis, what would you expect the mean μ_0 and standard deviation σ_0 of the data to be (in other words, what would the dataset look like if the baseline was true)?
- (c) Compute the z-score for the $n = 2420$ case, which is given by

$$z = \frac{\hat{p} - p_0}{\sigma_0}$$

Explain what the z-score measures.

- (d) Using a Z-test, compute the one-sided p-value by integrating directly. Explain what this p-value measures.
- (e) Repeat the calculations in (b) to (d) for the $n = 242$ high-confidence responses.

2. In Figure 2a of Tinsley et al., 2016, the authors use a horizontal dashed line to depict the upper limit of performance of an ideal detector. They calculate this upper limit using signal detection theory. We will replicate their calculations.

We assume that noise (no stimulus photons) and noise + signal are described via Poisson distributions. If the only source of photon noise are Poisson fluctuations, the probability of detecting n noise events is

$$p_d(n) = \frac{d^n}{n!} e^{-d}$$

Where d denotes the mean amount of noise events per time interval. In other words, letting X denote the number of photon-like detection events, $X \sim \text{Poisson}(d)$ for noise-only. For noise + signal, $X \sim \text{Poisson}(S)$, where $S = d + \eta E$. Here, E is the mean number of photons at the cornea and η the quantum efficiency of the detection system.

Now, recall the 2AFC task: In each trial, there are two time intervals. The photon, if present, is equally likely to be in interval 1 or interval 2. Let X_1 denote the photon-like event in interval 1, and X_2 the photon-like event in interval 2.

- (a) If the signal (photon) is in interval 1, what are the distributions of X_1 and X_2 (call this hypothesis H_1)? If the signal is in interval 2, what are the distributions of X_1 and X_2 (call this hypothesis H_2)?

Signal detection theory (SDT) assumes that the detector's decision in the 2AFC task depends on the likelihood ratio, given by

$$L(x_1, x_2) = \frac{f(x_1, x_2; H_1)}{f(x_1, x_2; H_2)}$$

Where x_1 and x_2 are the numbers of events in the first and second interval respectively. Here, $f(x_1, x_2; H_1)$ denotes the probability that conditioned on the signal being in interval 1 (conditioned on hypothesis H_1), there are x_1 detection events in interval 1 and x_2 detection events in interval 2.

- (b) Write a formula (using the Poisson distributions in (a)) for $f(x_1, x_2; H_1)$ and $f(x_1, x_2; H_2)$.
- (c) Show that the log-likelihood ratio can be written as

$$\ln L = (x_1 - x_2) \ln \left(\frac{S}{d} \right)$$

where $S = d + E\eta$ where E in this case is 1 photon.

- (d) Explain why the sign of $\ln L$ only depends on the sign of $x_1 - x_2$.

- (e) The detector must make a decision as to whether the detection event was in interval 1 (accept hypothesis H_1) or in interval 2 (accept hypothesis H_2). A simple rule is to choose H_1 if $L > 1$, and choose H_2 if $L < 1$. What hypothesis should the detector choose if $x_1 > x_2$? What about if $x_2 > x_1$? What about if $x_1 = x_2$ (note that in this case, both H_1 and H_2 are equally likely)?
- (f) Let $\Delta = X_1 - X_2$. Explain why the probability that a detector (following the above strategy) is correct, conditioned on the photon being in interval 1 (i.e. H_1 is true), is given by

$$P_{\text{correct}} = P(\Delta > 0) + \frac{1}{2}P(\Delta = 0)$$

Note that $P_{\text{correct}} \approx P(\Delta > 0)$. Assume that $\Delta \sim \mathcal{N}(\mu, \sigma^2)$, where $\mu = S - d$ and $\sigma^2 = S + d$. Let Φ denote the standard normal cumulative distribution function. Then, $P(\Delta > 0) = \Phi(\frac{\mu}{\sigma})$ (you don't need to show this).

- (g) The function Φ can be Taylor expanded to first order

$$\Phi(x) \approx 1 + \frac{x}{\sqrt{2\pi}}$$

Using this, provide an approximate formula for P_{correct} . Using $d = 1$ and $S = 1.1$, what (approximately) is P_{correct} ? This should be a good estimate of the theoretical performance limit of your ideal detector.

3. Activated rhodopsins are shut off after accumulating n successive phosphorylation events. Assume phosphorylation events arrive as a homogeneous Poisson process with rate λ (events/s).

Parameters: Set $\lambda = 200$ events/s, total molecules $M = 1000$, total time step count $N = 20,000$, and time step size $\Delta t = 1/N$ s = 5×10^{-5} s. The per-step expected number of events is $\lambda \Delta t = \lambda/N = 0.01$.

Simulation procedure: For each molecule $m = 1, \dots, M$:

- (a) Initialize time step $i = 0$ and event count $c = 0$.
- (b) For $i = 1, 2, \dots$:
 - (Simple approximation) Draw $X_i \sim \text{Bernoulli}(p)$ with $p = \lambda \Delta t = 0.01$ and set $c = c + X_i$; *or* (more exact) draw $X_i \sim \text{Poisson}(\lambda \Delta t)$ and set $c = c + X_i$ (allows multiple events within a step).
 - Whenever c reaches $k \in \{1, 3, 9, 15\}$ for the first time, record the step index $i_k^{(m)}$ (if multiple events occur in one step, assign all needed events to the current step).
 - Stop stepping for this molecule once $c \geq 15$ (i.e., after recording the 15th event time).

For molecule m , the time to the k th event is $T_k^{(m)} = i_k^{(m)} \Delta t$. Using the above simulation, do the following tasks:

- (a) Plot histograms of $\{T_k^{(m)}\}_{m=1}^M$ for $k = 1, 3, 9, 15$. Use the same binning across k when possible, and label axes (time in seconds, count or density).
- (b) For each $k = 1, 3, 9, 15$, report the sample mean $\hat{\mu}_k$, sample standard deviation $\widehat{\text{SD}}_k$, and coefficient of variation $\widehat{\text{CV}}_k = \widehat{\text{SD}}_k / \hat{\mu}_k$.

Some implementation notes:

- If you use the Bernoulli(p) per-step model, note it forbids more than one event in a single step; this is a close approximation here because $\lambda\Delta t = 0.01$ is small (the probability of ≥ 2 events in a step under Poisson(0.01) is $\approx 5 \times 10^{-5}$).